Appendix A

Experimental Design for Measuring Structurally Avoidable Energy Use

in Generative AI Inference

Status: Technical Appendix (Non-normative)
Purpose: To propose *one possible* experimental framework for validating or falsifying claims of structurally avoidable energy consumption during AI inference.
Scope: Inference stage only; no model retraining or weight modification.

A1. Research Question

Can measurable portions of inference-stage energy consumption be attributed to **structural design choices**, rather than model capability, and therefore reduced **without changing model parameters**?

A2. Measurement Targets

The following metrics are recommended as primary observables:

1. **Energy per request (J/request)**
2. **Energy per token (J/token)**
3. **Total inference energy (J) per response**
4. **Latency (ms)** (secondary; used to detect hidden retries)

Energy should be computed as:

E= ∫ t0t1P(t) dtE = \int_{t_0}^{t_1} P(t) \, dtE= ∫ t0t1P(t)dt

where P(t)P(t)P(t) is instantaneous power draw during inference execution.

A3. Experimental Controls and Variants

Experiments should follow a **single-variable perturbation principle**.
Each variant modifies *only one structural component* at a time.

A3.1 Control Condition

- Default inference pipeline
- Default output verbosity and safety handling
- Default multi-path or hedging behavior
- No explicit state or time injection beyond baseline system design

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A3.2 Variant Group A — Output Redundancy

Objective: Measure energy impact of non-essential output tokens.

- **Variant A1 (Compressed Output):**
  Remove fixed introductory phrases, politeness markers, and stylistic padding while preserving semantic correctness.
- **Measurement:**
  Compare EcontrolE_{control}Econtrol vs EA1E_{A1}EA1
  Normalize by token count.

Falsification criterion:
If ΔE/Econtrol<5%\Delta E / E_{control} < 5\%ΔE/Econtrol<5%, output redundancy is likely overstated.

A3.3 Variant Group B — Hedging / Multi-Path Inference

Objective: Quantify energy overhead from generating and discarding intermediate candidates.

- Variant B1 (Single-path inference):
  Disable candidate branching, self-correction loops, or retry logic where technically feasible.
- Measurement:
  Compare:
  - Number of internal generations
  - Total energy per response

Falsification criterion:
If single-path inference does not materially reduce energy consumption, hedging loops are not a dominant contributor.

A3.4 Variant Group C — State and Time Handling

Objective: Compare inference-based state approximation with direct state access.

- Variant C1 (Direct state access):
  Provide system time, session state, or mode information via low-cost system calls.
- Variant C2 (Inference-based approximation):
  Force the model to infer state implicitly.
- Measurement:
  Compare energy cost of C1 vs C2 under identical prompts.

Falsification criterion:
If state injection does not reduce inference energy, state-approximation overhead is negligible.

A4. Instrumentation Recommendations

The following tools are sufficient for reproducible measurement:

- GPU-level power:
  - `nvidia-smi`
  - IPMI sensors
- CPU-level power:
  - Intel RAPL

- **System-level power:**
    - Smart PDU / wall power meter

Sampling frequency should be sufficient to capture short inference bursts (≥10 Hz recommended).

A5. Experimental Design Notes

- Each run should be repeated **N ≥ 30** times to control for variance.
- Prompts should be held constant across control and variant runs.
- Background system load must be monitored or isolated.

A6. Expected Outcome Space (Non-prescriptive)

This framework does **not** assume a specific outcome. Possible results include:

- **Partial validation:**
  One or two structural categories show significant energy savings.
- **Full falsification:**
  No statistically significant differences observed.
- **Mixed results:**
  Energy reduction observed but below hypothesized ranges.

All outcomes are considered valid contributions.

A7. Interpretation Boundary

- Results should **not** be generalized beyond tested configurations.
- Percentages refer to **structural deltas under experimental conditions**, not global averages.
- This appendix proposes *a test*, not *a conclusion*.

A8. Reproducibility Statement

This experimental design is intended to be:

- Model-agnostic
- Hardware-flexible
- Reproducible by independent labs

Disagreement with results is explicitly welcomed, provided measurements are disclosed.

Appendix Summary

This appendix provides a **minimal, falsifiable experimental scaffold** for evaluating whether inference-stage energy use contains structurally avoidable components.
It does not argue *that* such inefficiencies exist at scale—only that they are **measurable, testable, and worth quantifying**.